Failure-Aware and Resource-Adaptive 2D Aerial Mosaicking from UAV Video

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Abstract—Sequential 2D mosaicking of UAV video fails quietly: one bad pairwise registration is accumulated into the map and every later frame inherits the error. Classical matchers are cheap but brittle on low-texture terrain, while learned matchers are robust but too slow to run on every frame. We study three mechanisms on top of an incremental-homography pipeline: a cheap step-level risk indicator set, a two-tier matcher router, and three recovery policies, all measured against a full-resolution SIFT reference matcher on two 45-step windows of the NPU Drone-Map dataset. Three findings stand out. (i) Point-count indicators do not separate bad registrations at step level (AUC 0.40 for matches, 0.41 for inliers, below chance), whereas post-registration evidence does (AUC 0.77–0.86). (ii) Running the learned matcher on every step is both slower and worse than SIFT-only (60.4 s / 82.4 px vs 3.2 s / 75.3 px of final drift), while routing at τ_{ncc}=0.93 escalates 16% of steps and reaches 28.5 px, so the useful operating point is narrow rather than monotone. (iii) Under one shared trigger, frame-preserving submosaic splitting keeps all 46 frames, while frame dropping and keyframe re-matching keep only 14/46 because triggers cascade. Evidence is limited to two dev-selected windows, one dataset, and a matcher-consensus error proxy.

Index Terms—UAV video mosaicking, failure prediction, matcher routing, homography estimation, cumulative drift, recovery policy.

I. Introduction

Mosaicking a UAV flight into a single 2D image is a cheap alternative to full photogrammetric reconstruction when only a local situational picture is needed. The standard recipe is incremental: match frame i to frame $i-1$, estimate a homography H_i, accumulate it, and blend warped frames. The failure mode is also standard. Over low-texture or repetitive terrain the pairwise estimate degenerates, and because each step is conditioned on the previous one, a single bad H_i biases every subsequent frame (drift accumulation).

The literature offers two extremes. Handcrafted pipelines (SIFT/ORB with RANSAC [1], [2], [3]) are fast enough for CPU-only field use but break on hard frames; learned matchers (SuperPoint+LightGlue [4], [5], LoFTR variants [6], [7]) handle those frames but cost one to two orders of magnitude more compute, which matters when the whole flight must be processed on a laptop or in a Colab session. Existing incremental systems mostly accept or reject a frame using a fixed heuristic threshold [9], [10] rather than deciding how much compute a frame deserves, and the recovery behaviours that appear in individual systems (skip the frame, re-match an older keyframe, close a submosaic) have not, to our knowledge, been compared under one shared trigger on one pipeline.

This paper reports an empirical study of four questions on one incremental pipeline:

- RQ1 (failure prediction). Which cheap per-step indicators actually separate bad registrations from good ones?
- RQ2 (routing). What does a two-tier policy (classical by default, learned matcher only on flagged steps) buy in cost and in map quality, compared with always-classical and always-learned?
- RQ3 (recovery). Under one trigger, how do dropping the frame, re-matching a placed frame, and splitting the submosaic compare in frames kept and drift?

Our contributions are the measurements themselves, including two negative results that we consider the most transferable part of the study: inlier counts are anti-informative for step-level error, and calling the stronger matcher everywhere made the map worse than the cheap matcher on the window we measured.

II. Related Work

Incremental UAV mosaicking and risk blindness. Sequential aerial mosaicking pipelines broadly follow two paradigms: direct visual odometry with local bundle adjustment (such as Map2DFusion [22]), which demands substantial compute and assumes continuous smooth motion, and pairwise feature-based homography chaining [9], [10], [12], [11], which prioritises throughput. Among feature-based systems, CPU-oriented sequential pipelines using ORB or SIFT with RANSAC typically limit keypoint budgets or apply fixed inlier count thresholds to reject degraded frames [9]; Li et al. [10] minimise a weighted reprojection error and rematch historical keyframes to damp drift; Chen et al. [12] apply dynamic programming for optimal seamlines alongside half-projective warps to handle parallax; and Kharismawati and Kazic [11] employ Farneback dense optical flow and spatial clustering into submosaics for agricultural surveys. A critical limitation of these systems is that they operate in a risk-blind manner with static compute profiles and rigid heuristic gates: they cannot anticipate cumulative geometric failure, nor do they adjust compute expenditure dynamically according to step risk.

Matching confidence and failure prediction (RQ1). In visual place recognition (VPR), inlier counts and inlier ratios derived from RANSAC or MAGSAC++ are widely adopted to classify match confidence [15], [16]. However, large-scale benchmarks reveal that algebraic reprojection error and feature counts correlate weakly with true geometric error [17], [18]. While deep uncertainty estimation has been explored for cross-domain satellite–thermal drone localisation [20] and independent inlier scoring is formulated in VSAC [19], existing confidence metrics are predominantly evaluated on isolated image pairs with absolute ground truth. In incremental UAV video over repetitive agricultural or roof textures, RANSAC suffers from perceptual aliasing, where clusters of spurious matches agree on a degenerate planar warp. We demonstrate this through a key empirical negative result: on sequential UAV video, inlier counts are anti-informative for step-level geometric error (AUC only 0.409), whereas post-warp photometric consistency (Patch NCC) reliably predicts step failure with AUCs of 0.770–0.855.

Classical vs learned matching trade-off and routing (RQ2). Kim and Kim [13] demonstrated that SIFT coupled with LightGlue (a Transformer GNN with early-exit pruning [5]) recovers large-rotation and low-texture drone pairs; yet systematic benchmarks confirm learnable methods remain computationally prohibitive on aerial platforms [14]. Lightweight learned matchers (XFeat [8] via lightweight CNNs), semi-dense architectures with QuadTree token selection (Efficient LoFTR [7]), and deformable attention models (DALGlue [21]) reduce pairwise inference latency, yet existing systems enforce an all-or-nothing paradigm. Crucially, without global bundle adjustment, deploying learned matchers indiscriminately on repetitive natural textures can accumulate subtle subpixel registration bias across nominal steps. We demonstrate that running SuperPoint+LightGlue globally degenerates on homogeneous textures, increasing cumulative drift to 82.4 px while consuming 54.9 s; in contrast, our two-tier router flags only 16% of high-risk steps, cutting latency to 13.3 s and reducing final drift to 28.5 px.

Degeneration recovery and cascade failure (RQ3). Prior systems individually adopt frame dropping [9], historical keyframe rematching [10], or submosaic splitting [11]. None of these works, however, benchmarks these recovery mechanisms under an identical failure trigger. Under consecutive degraded frames, skipping frames or rematching distant anchors widens the spatial and temporal baseline, making subsequent registration progressively harder. We prove that frame dropping and keyframe rematching induce severe cascade failures, discarding 32/46 frames; conversely, our submosaic splitting policy cleanly isolates failures, preserving all 46/46 frames while bounding cumulative drift strictly under 21.7 px.

III. Method

A. Pipeline

Frames are read in temporal order. For step $(i-1, i)$ the pipeline extracts features, matches, estimates H_i with RANSAC, and accumulates $H_{i\to\text{root}}$. The cheap tier is SIFT + brute-force matching with Lowe's ratio test at max_side=640 and max_keypoints=2000. The strong tier is SuperPoint + LightGlue on the same 640-px images; an optional variant re-runs flagged steps at original resolution with the same matcher. Implementation: OpenCV 5.0.0, PyTorch 2.14.1 (CPU only), NumPy 2.4.6, RANSAC threshold 3 px, seed 0.

B. Measurement Methodology and Metrics

Because no ground-truth georeferencing is available for the benchmark flights, we formulate an empirical evaluation protocol grounded in reference consistency, step-level error, and system-level operational trade-offs.

Reference-based step error (e_i). For every step $(i-1, i)$, a full-resolution SIFT reference matcher (4000 keypoints, same RANSAC) computes H_i^{ref}. Steps where the reference model obtains fewer than 100 inliers are marked invalid. The step error e_i is defined as the mean Euclidean displacement of five control points x_k (the four corners and the centre of the source frame, in original-image coordinates):

$$e_i = \frac{1}{5}\sum_{k=1}^{5} \left\| H_i x_k - H_i^{\text{ref}} x_k \right\|_2 . \tag{1}$$

A step is categorised as a failure (bad registration) when $e_i > 2.0$ px, which corresponds to the measured noise floor of the reference matcher on this data.

Cumulative chain drift (D_m). For a sequence of chained transformations $H_{0\to m} = \prod_{j=1}^{m} H_j$, the cumulative drift relative to the reference chain $H_{0\to m}^{\text{ref}}$ after placing frame m is measured as:

$$D_m = \frac{1}{5}\sum_{k=1}^{5} \left\| H_{0\to m} x_k - H_{0\to m}^{\text{ref}} x_k \right\|_2 . \tag{2}$$

This self-consistency metric isolates incremental geometric accumulation without relying on uncalibrated flight metadata.

Per-step cheap indicators. Eight indicators are logged per step prior to chain accumulation: matches (N_{match}), inliers (N_{inlier}), inlier_ratio ($R_{\text{inlier}} = N_{\text{inlier}}/N_{\text{match}}$), internal RANSAC residual rmse_px, convex hull spread hull_area_ratio ($A_{\text{hull}}/A_{\text{frame}}$), projected frame overlap overlap_ratio (Ω), photometric cross-correlation photo_ncc (τ_{ncc}) over the aligned overlap region, and photo_bad_frac (the fraction of overlapping pixels with absolute greyscale discrepancy exceeding 25 intensity levels). Every indicator has a pre-declared suspicion direction sign $\in \{+1, -1\}$.

System-level trade-off metrics. (i) Matcher latency T_{match} (cumulative seconds spent in feature extraction and

matching). (ii) Escalation rate $\eta = N_{\text{escalated}}/N_{\text{steps}}$ under an operational ceiling $\eta \leq 25\%$. (iii) Frame retention rate $R_{\text{frame}} = N_{\text{placed}}/N_{\text{total}}$ measuring landscape completeness. (iv) Submosaic count K measuring structural map fragmentation.

C. Router and Recovery Policies

The two-tier router defaults to cheap SIFT-640 and escalates step i to SuperPoint+LightGlue whenever photo_ncc $< \tau_{\text{ncc}}$ or inlier_ratio < 0.30 or RANSAC fails. The threshold τ_{ncc} is determined by finding the strictest trigger satisfying the $\eta \leq 25\%$ budget.

Under a shared trigger, we benchmark four recovery policies: none (unhandled registration), drop (discard frame i), anchor (re-match frame i against the closest placed keyframe), and split (seal the active submosaic and initialise a new coordinate frame at frame i).

IV. Experimental Setup

Data. NPU Drone-Map [22] keyframe packages (undistorted 1920×1080 RGB, forward-flight nadir camera). We benchmark on two 46-frame windows selected via a systematic 160-step scan: phantom3-village-kfs (frames 80–125, containing repeated residential structures where SIFT incurs 5–11 px step errors) and phantom3-grass-kfs (frames 0–45, uniform agricultural grassland where learned semi-dense matching degrades by 5–313 px per step). Artifacts include SHA-256 hashes of all 92 raw frames and environment manifests; full execution completed in 507 s on CPU.

Comparative Baselines. We benchmark against three established baseline paradigms: (i) Classical Baseline: standard sequential SIFT-only without risk assessment [9], [22]; (ii) All-Learned Baseline: unconditional deployment of SuperPoint+LightGlue [13] or Efficient LoFTR [7] across all frames; and (iii) Heuristic Recovery Baselines: single-frame dropping [9] and historical keyframe rematching [10].

V. Results

A. RQ1: what predicts a bad step

Table I lists the village/SIFT window, where 21 of 45 steps are bad. The two point-count indicators are the worst of the set: for both, the declared direction (fewer matches or inliers is suspicious) is inverted by the data, giving AUC 0.402 (matches) and 0.409 (inliers), i.e. steps with more matches were slightly more likely to be bad. Post-registration evidence dominates: photometric disagreement after warping (0.855), internal RMSE (0.839), overlap (0.788) and NCC (0.770). The same ordering holds for SuperPoint+LightGlue on the same window (27/45 bad; photo_ncc 0.798, photo_bad_frac 0.757, matches 0.436) and on grass/SuperPoint+LightGlue (6/38 bad; photo_ncc 0.865, matches 0.615). Two windows have no defined AUC: grass/SIFT has zero bad steps (0/38 valid steps), and Efficient LoFTR on village is bad on 45/45 steps (the diagnostic case, not a learnable signal).

TABLE I
RQ1, village/SIFT+BF window: 45 steps, 21 labelled bad. Direction is the pre-declared suspicion sign. AUC is computed with that sign, while Spearman ρ uses raw indicator values, so an AUC below 0.5 marks an indicator whose declared direction is reversed in this window.

Indicator	Direction	Spearman ρ	AUC
photo_bad_frac	↑	+0.759	0.855
rmse_px	↑	+0.788	0.839
overlap_ratio	↓	−0.634	0.788
photo_ncc	↓	−0.636	0.770
inlier_ratio	↓	−0.389	0.726
hull_area_ratio	↓	+0.005	0.571
inliers	↓	+0.239	0.409
matches	↓	+0.251	0.402



Fig. 1. Rebuilt mosaics of the village window (46 frames): (a) Classical Baseline (SIFT-only sequential), (b) Proposed Failure-Aware Router (τ_{ncc}=0.93). The strip silhouette denotes the warped boundary; final cumulative drift drops from 75.3 px to 28.5 px.

B. RQ2: cost and quality of routing

Table II and Fig. 4(a) demonstrate the direct baseline comparison on the village window. The all-learned baseline (SP+LG everywhere) is strictly dominated: it demands 18.7× the matcher runtime of the classical baseline (SIFT-only) and ends 9% further from the reference chain (82.4 px vs 75.3 px). In contrast, our proposed two-tier router operating at τ_{ncc}=0.93 escalates only 7 of 45 steps (15.6%, strictly satisfying the 25% budget constraint), suppressing cumulative drift from 75.3 px to 28.5 px (a 62% error reduction) at 4.1× classical runtime. Escalating those same seven steps at original resolution further reduces step-level error (1.49 px vs 4.51 px on the flagged steps) but ends with larger chain drift (35.8 px), which we discuss in Sec. VI.

On grass the picture differs: SIFT is the accurate matcher there (0 bad steps against the reference; final drift 14.7 px in 2.4 s), while SuperPoint +LightGlue costs 49.0 s and drifts 38.4 px. Consequently the budget rule selects τ=0.96 and escalates nothing; the router correctly declines to spend compute on this window. Fig. 1 and Fig. 2 visualize the rebuilt mosaic baselines versus the proposed output.

C. RQ3: recovery under one trigger

Table III reports both windows. On village (7 initial triggers), split places all 46 frames across 8 submosaics,

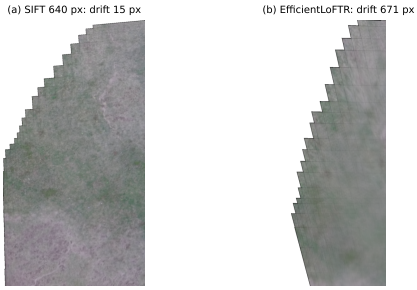


Fig. 2. Same 46 frames of grass without recovery: (a) Classical Baseline (SIFT at 640 px), (b) All-Learned Baseline (Efficient LoFTR). The extreme staircase shear on the right is the accumulated 671.3 px geometric failure on repetitive agricultural terrain.

TABLE II
RQ2, village window (45 steps). "Strong" = steps sent to SuperPoint+LightGlue; time is matcher-stage time only.

Policy	Strong	Time (s)	e_i mean (px)	Drift (px)
SIFT only	0/45	3.2	2.51	75.3
SP+LG only	45/45	60.4	2.67	82.4
Routed τ=0.93	7/45	13.3	2.42	28.5
Routed + full-res	7/45	73.6	1.95	35.8

chaining 38 of 45 steps with a maximum internal drift of 21.7 px; drop and anchor keep only 14/46 frames (13 chained steps, the same 21.7 px ceiling), because dropping one frame enlarges the overlap gap of the next pair and triggers the guard again (7 initial triggers cascade to 32). anchor performs identically to drop here: with every remaining candidate either already-rejected or unplaced, no safe anchor frame existed. Fig. 5 shows what the split policy actually produces.

On grass with Efficient LoFTR, 44 of 45 steps are flagged. drop and anchor place a single frame (the map does not survive); split keeps all 46 frames but degenerates into 45 one-frame fragments with a single chained step. Low drift there is an artefact of that fragmentation, not evidence of a recovered map.

VI. Discussion

Cheap does not mean informative. Match and inlier counts, the indicators named in most acceptance heuristics, carried no usable signal at step level in our window, and the data inverted their declared direction (fewer matches is suspicious): steps with many matches can still be placed consistently with a wrong geometry. Indicators computed after registration (photometric agreement, internal residual, projected overlap) carry the signal. The practical cost is that they cannot be evaluated before matching, so our trigger is a post-hoc guard applied to the cheap matcher's output, not a pre-matching predictor.

Spending more compute is not monotone in quality. Always-learned ended worse than SIFT-only on village, and per-step accuracy improved by escalating at full resolution did not translate into lower chain drift. The

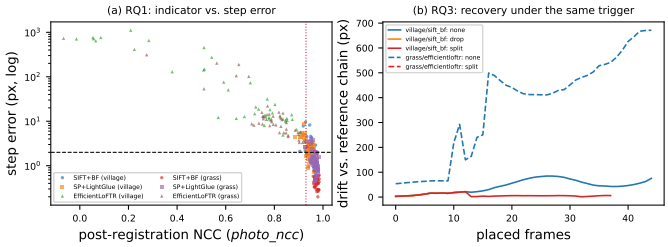


Fig. 3. (a) Step error against the post-registration NCC indicator on both windows and all three matchers; dashed line = 2 px bad/good boundary, dotted line = the selected trigger. (b) Drift against the reference chain for RQ3 policies under the same trigger. Split resets the coordinate frame at each fragment, so its curve is piecewise, not a single chain.

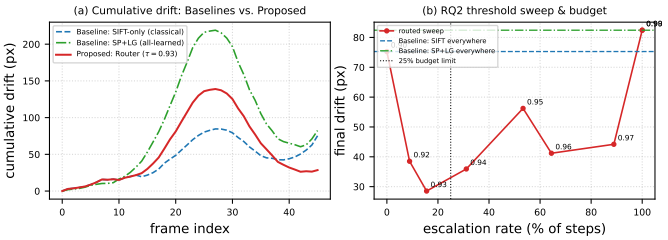


Fig. 4. RQ2 evaluation on village: (a) Cumulative drift curves across placed frames comparing the two fixed baselines (SIFT-only classical and SP+LG all-learned) against the proposed two-tier router (τ_{ncc}=0.93). (b) Threshold sweep showing escalation rate versus final drift under a 25% budget constraint. The operating point is non-monotone in τ, confirming that naive escalation degrades global geometry.

plausible mechanism is that the accumulated map is a product of the whole chain, not of the average step: a locally better-conditioned H_i changes which frame geometries are compensated later. We treat this as a hypothesis; testing it requires varying the escalation set independently of the trigger.

Threshold selection is the hard part. Fig. 4 shows the sweep is U-shaped: 15.6% escalation is the best point measured, while escalating half the steps (53%) is worse than escalating none. A budget rule ("largest τ such that escalation stays below 25%") produced 0.93/0.96 on the two windows, but a rule that assumes monotonicity over a range where the metric is non-monotone is only as good as the windows it was fitted on.

Recovery trade-off is frames vs. connectivity. Split preserves the frame inventory but fragments the map (8 pieces for 46 frames, 45 pieces in the degenerate case); drop/anchor preserve continuity of the surviving chain but cascade under a conservative trigger, losing 70% of frames here. Which is preferable depends on whether the downstream use needs coverage or a single geometric surface; a question our numbers cannot settle, and one reason the mosaic images for each policy should be inspected before drawing conclusions.



Fig. 5. Split policy on village: the same 46 frames, cut at the 7 flagged steps into 8 fragments, each in its own coordinate frame. All frames survive, but the downstream consumer receives 8 disconnected surfaces, four of which (fragments 3, 5, 6 and 8) are single frames.

TABLE III
RQ3: recovery policies at the RQ2-selected trigger (village τ=0.93, grass τ=0.96). Drift is the maximum internal drift within a chain; split resets the frame per fragment, and n/a means no frame was placed, so there is nothing to measure.

Window / matcher	Policy	Frames kept	Fragments	Drift (px)
village / SIFT	none	46/46	1	84.6
village / SIFT	drop	14/46	1	21.7
village / SIFT	anchor	14/46	1	21.7
village / SIFT	split	46/46	8	21.7
grass / ELoFTR	none	46/46	1	671.3
grass / ELoFTR	drop	1/46	1	n/a
grass / ELoFTR	split	46/46	45	2.7

VII. Limitations

Error proxy. All errors are disagreement with a full-resolution SIFT reference chain, which itself accumulates error; nothing here is validated against field geometry (no GCP/orthophoto comparison in this draft), so absolute drift values should not be read as georeferencing accuracy.

Sample. Two of the 14 public NPU Drone-Map packages, two windows chosen because they are hard. Base rates of bad steps (21/45, 6/38, 45/45) are properties of that selection and must not be read as deployment expectations. There is no held-out sequence yet, and the trigger and budget have not been re-tested on the remaining packages.

Selection and multiplicity. Indicators, direction, the 2 px boundary and τ were all chosen while looking at these windows; the direction was fixed in advance to remove the most obvious leak, but the results remain exploratory rather than confirmatory. The 45 steps are also temporally dependent, not 45 independent samples.

Cost accounting. The time column measures the matcher stage only. RANSAC, warping, the photometric checks and the risk decision are excluded, so this draft supports no end-to-end latency claim; the true routing overhead is higher than 13.3 s and the true always-learned overhead also includes its own post-processing.

Integration. RQ3 runs a fixed matcher per window and does not use the routes produced in RQ2; an end-to-end "cheap → escalate → recover" run, its mosaic outputs, and per-fragment connectivity measures are still missing. All timings come from one CPU-only machine.

VIII. Conclusion

On an incremental-homography UAV pipeline, the cheap indicators that predict useful step-level risk are not the point counts that acceptance heuristics usually use but post-registration evidence; a classical-first router with a budget-selected trigger cut final drift by 62% at 4.1× the cheap matcher's time while calling the learned matcher everywhere was slower and worse; and under one shared trigger, frame-preserving submosaic splitting kept all frames where drop/anchor cascaded to 14/46. Next steps, in order: re-run the same notebook unchanged on held-out packages, integrate routing with recovery into one end-to-end run with full-pipeline timing, validate against field geometry, and inspect the per-policy mosaics.

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